

Diversity and trophic structure characterize biological communities

Ecological communities can be characterized by certain general attributes, including how diverse they are and the feeding relationships of their species. In some cases, as you'll read, a few species exert strong control on a community's *structure*—the number, identity, and relative abundance of its species.

Species Diversity

The **species diversity** of a community—the variety of different kinds of organisms that make up the community—has two components. One is **species richness**, the number of different species in the community. The other is the **relative abundance** of the different species, the proportion each species represents of all individuals in the community.

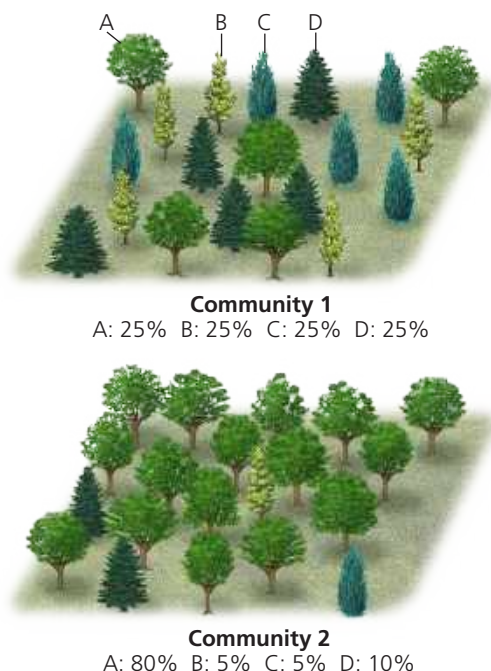
Imagine two small forest communities, each with 100 individuals distributed among four tree species (A, B, C, and D) as follows:

Community 1: 25A, 25B, 25C, 25D

Community 2: 80A, 5B, 5C, 10D

The species richness is the same for both communities because they both contain four species of trees, but the relative abundance is very different (**Figure 54.12**). You would easily notice the four types of trees in community 1, but

▼ **Figure 54.12 Which forest is more diverse?** Ecologists would say that community 1 has greater species diversity, a measure that includes both species richness and relative abundance.



unless you looked carefully, you might see only the abundant species A in the second forest. Most observers would intuitively describe community 1 as the more diverse of the two communities.

Ecologists use many tools to compare the diversity of communities. They often calculate indexes of diversity based on species richness and relative abundance. One widely used index is the **Shannon diversity index** (H):

$$H = -(p_A \ln p_A + p_B \ln p_B + p_C \ln p_C + \dots)$$

where A, B, C ... are the species in the community, p is the relative abundance of each species, and $\ln$ is the natural logarithm; the $\ln$ of each value of p can be determined using the "ln" key on a scientific calculator. A higher value of H indicates a more diverse community. Let's use this equation to calculate the Shannon diversity index of the two communities in Figure 54.12. For community 1, $p = 0.25$ for each species, so

$$H = -4(0.25 \ln 0.25) = 1.39.$$

For community 2,

$$H = -[0.8 \ln 0.8 + 2(0.05 \ln 0.05) + 0.1 \ln 0.1] = 0.71.$$

These calculations confirm our intuitive description of community 1 as more diverse.

Determining the number and relative abundance of species in a community can be challenging.

Because most species in a community are relatively rare, it may be hard to obtain a sample size large enough to be representative. It can also be difficult to identify some of the species in the community. If an unknown organism cannot be identified on the basis of morphology alone, it is useful to compare all or part of its genome to a reference database of DNA sequences from known organisms. For example, although the two samples of red algae shown in **Figure 54.13** might appear to be two different species, comparing their sequences of a short standardized section of DNA—a DNA "barcode"—to a reference database shows that they belong to the same species. Researchers are increasingly using DNA sequencing for species identification as it becomes cheaper and as DNA sequences from more organisms are added to comparative databases.

It can also be difficult to census the highly mobile or less visible members of communities, such as microorganisms, deep-sea creatures, and nocturnal species. The small size of microorganisms makes them particularly difficult to sample, so ecologists now commonly use molecular tools to help determine microbial diversity (**Figure 54.14**).

▼ **Figure 54.13** Two samples of one species, the red alga *Chondracanthus harveyanus*.

